AI SL — Specific Topics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(5)$.

Final Answer: 23. Substitute $x = 5$: $f(5) = 4(5) + 3 = 23$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 5(2) + 4 = 14$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 86°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 45^\circ) = 86^\circ$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

1. Substitute $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$. When $x = 0$, $\theta = 0$. When $x = 1$, $2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4 - x^2} = \sqrt{4 - 4\sin^2\theta} = 2\sqrt{\cos^2\theta} = 2\cos\theta$ (since $\cos\theta > 0$ on $[0, \pi/6]$). 3. Substitute into the integral.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.

1. Sort all edges in non-decreasing order of weight: DE (1), BC (2), AB (3), CD (4), BD (5), CE (6), AC (7), AD (8). 2. Select edges avoiding cycles: - Add DE (weight 1) [Vertices connected: {D, E}] - Add BC (weight 2) [Vertices connected: {D, E}, {B, C}] - Add AB (weight 3) [Vertices connected: {D, E}, {A, B, C}] - Add CD (weight 4) [Merges components: {A, B, C, D, E} - all 5 vertices connected!] 3. Check edge count: $V - 1 = 5 - 1 = 4$ edges selected. 4. Total MST Weight: $1 + 2 + 3 + 4 = 10$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

Kruskal's algorithm finds a global minimum spanning tree that connects all vertices in the graph with minimum total edge weight, without reference to any single starting vertex. Dijkstra's algorithm finds the shortest path routes from a designated single source vertex to all other individual vertices. A minimum spanning tree does not necessarily contain the shortest paths between arbitrary pairs of vertices.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

Use $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \ldots$ with $u = 2x$: $\ln(1+2x) = 2x - 2x^2 + \frac{8}{3}x^3 - \ldots$. Convergence requires $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[AA HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[AI HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi+2\pi k}{4}\right) + i\sin\left(\frac{\pi+2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right]$, $k \in \{0, 1, 2, 3\}$. Step 3: Evaluate for each k:$k = 0$.

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12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

1. Expand $(\cos\theta + i\sin\theta)^3$ via Binomial Theorem: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. 2. By De Moivre's Theorem, this equals $\cos 3\theta + i\sin 3\theta$. Equate real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$. Substitute $\sin^2\theta = 1 - \cos^2\theta$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x\,dx$ for integers $n \geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x\,dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x\,dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1}\,dx$. Let $dv = e^x\,dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1}e^x\,dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}$. ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e-1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e-2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

(a) Vertical Asymptote: The denominator is zero at $x = 3$. Numerator at $x = 3$ is $3^2 - 3 - 2 = 4eq0$. Thus $x = 3$ is a vertical asymptote. Oblique Asymptote: Perform polynomial long division: $\frac{x^2 - x - 2}{x - 3} = x + 2 + \frac{4}{x-3}$. As $x \to \pm\infty$, $\frac{4}{x-3} \to 0$. Therefore, $y = x + 2$ is the oblique asymptote. (b) Differentiate $f(x) = x + 2 + 4(x-3)^{-1}$: $f'(x) = 1 - \frac{4}{(x-3)^2}$. Set $f'(x) = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.01 Calculus**

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \leq \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 1.01 Number & Algebra**

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4\ \text{m}$. Initial gravitational potential energy: $\text{PE} = mgh = 5(9.8)(4) = 196\ \text{J}$ 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}\ \text{N}$. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}\ \text{N}$. Work done against friction over distance $d = 8\ \text{m}$: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896\ \text{J}$ 3. By.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.01 Functions**

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 Geometry & Trigonometry**

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Show that the sum of these five roots equals 0.

1. Write $-32 = 32e^{i\pi}$. By De Moivre's Theorem: $z_k = 32^{1/5}\exp\left(i\frac{\pi + 2\pi k}{5}\right) = 2e^{i(\pi + 2\pi k)/5}$ for $k = 0, 1, 2, 3, 4$. Evaluating the arguments in $(-\pi, \pi]$: - $k = 0$: $z_0 = 2e^{i\pi/5}$ - $k = 1$: $z_1 = 2e^{i3\pi/5}$ - $k = 2$: $z_2 = 2e^{i5\pi/5} = 2e^{i\pi} = -2$ - $k = 3$: $z_3 = 2e^{i7\pi/5} = 2e^{-i3\pi/5}$ - $k = 4$: $z_4 = 2e^{i9\pi/5} = 2e^{-i\pi/5}$ 2. Sum of the roots: Notice that.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 4.01 Statistics & Probability**

In a circle of radius $10\ \text{cm}$, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

(a) Use the cosine rule in triangle OAB with $OA = OB = 10$ and $\angle AOB = \frac{2\pi}{3} = 120^\circ$: $AB^2 = 10^2 + 10^2 - 2(10)(10)\cos\left(\frac{2\pi}{3}\right) = 100 + 100 - 200\left(-\frac{1}{2}\right) = 200 + 100 = 300$ Thus $AB = \sqrt{300} = 10\sqrt{3}\ \text{cm}$. (b) Area of sector OAB: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(10^2)\left(\frac{2\pi}{3}\right) = \frac{100\pi}{3}\ \text{cm}^2$ Area of triangle OAB.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Calculus

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

Base Step: For $n = 1$: $5^1 - 1 = 4 = 4(1)$. Since 4 is divisible by 4, the statement $P(1)$ is true. Inductive Hypothesis: Assume $P(k)$ is true for some positive integer $k \geq 1$, meaning:

$5^k - 1 = 4m$ for some integer $m \in \mathbb{Z} \Rightarrow 5^k = 4m + 1$. Inductive Step: Consider $n = k + 1$:

$5^{k+1} - 1 = 5 \cdot 5^k - 1$. Substitute the inductive hypothesis $5^k = 4m + 1$:

$5(4m + 1) - 1 = 20m + 5 - 1 = 20m + 4 = 4(5m + 1)$. Since $m \in \mathbb{Z}$, $5m + 1$ is an integer, so.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(2)$.

a. 12 ✓

b. 14

c. 4

d. 16

Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for amplitude, vertical shifts and dilations of trigonometric graphs consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solutions to trigonometric equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 5$ in linear regression line, evaluate $f(3)$.

a. 20 ✓

b. 25

c. 15

d. 24

Substitute $x = 3$: $f(3) = 5(3) + 5 = 20$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

$x2500isinvestedatanannualrateof8\%$, compounded monthly, for 3 years. Find the value of the investment, correct to 2 decimal places.

a. **3175.59** ✓

b. 3145.59

c. 3115.59

d. 3205.59

Working through this modelling step: $A = P(1 + \frac{r}{100k})^{kt} = (2500)(1 + \frac{8}{100 \times 12})^{12 \times 3} \approx 3175.59$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

$x2500isinvestedatanannualrateof6\%$, compounded quarterly, for 2 years. Find the value of the investment, correct to 2 decimal places.

a. 2756.23

b. 2846.23

c. **2816.23** ✓

d. 2786.23

Working through this modelling step: $A = P(1 + \frac{r}{100k})^{kt} = (2500)(1 + \frac{6}{100 \times 4})^{4 \times 2} \approx 2816.23$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

$x2000isinvestedatanannualrateof6\%$, compounded semi-annually, for 2 years. Find the value of the investment, correct to 2 decimal places.

a. 2191.02

b. **2251.02** ✓

c. 2281.02

d. 2221.02

Working through this modelling step: $A = P(1 + \frac{r}{100k})^{kt} = (2000)(1 + \frac{6}{100 \times 2})^{2 \times 2} \approx 2251.02$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A car purchased for $x20000depreciatesatarateof20\%$ per year (reducing balance). Find its value after 3 years, correct to 2 decimal places.

a. **10240.00** ✓

b. 9640.00

c. 10540.00

d. 9940.00

Working through this modelling step: $A = P(1 - \frac{r}{100})^{n} = (20000)(1 - \frac{20}{100})^{3} \approx 10240.00$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

$x1500isinvestedatacompoundinterestrateof7\%$ per year, compounded annually, for 5 years. Find the value of the investment, correct to 2 decimal places.

a. **2103.83** ✓

b. 2063.83

c. 2143.83

d. 2023.83

Working through this modelling step: $A = P(1 + \frac{r}{100})^{n} = (1500)(1 + \frac{7}{100})^{5} \approx 2103.83$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Site $A(1, -1)$ and site $B(8, 1)$ generate a Voronoi diagram. Determine which site the point $P(5, -1)$ is closer to.

a. Site B ✓

b. Site A

c. Equidistant from A and B

d. Cannot be determined

$AP^2 = (5-1)^2 + (-1+1)^2 = 16$, $BP^2 = (5-8)^2 + (-1-1)^2 = 13$. Since $13 < 16$, P is closer to site B.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

$x1200isinvestedatasimpleinterestrateof8\%$ per year for 5 years. Find the interest earned.

a. 480.00 ✓

b. 470.00

c. 490.00

d. 500.00

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(1200)(8)(5)}{100} = 480.00$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given $\log_5(6) = p$, express $\log_5(6^3)$ in terms of p.

a. $p + 3$

b. $3p$ ✓

c. $\frac{p}{3}$

d. p^3

Working through this algebra step: By the log law $\log(x^k) = k\log x$, the expression equals $3p$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve for x, correct to 2 decimal places: $2^{x+2} = 41$.

a. 2.86

b. 2.36

c. 3.36 ✓

d. 3.86

Working through this algebra step: $x + 2 = \log_2(41) \Rightarrow x = \log_2(41) - 2 \approx 3.36$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert $\frac{\pi}{2}$ radians to degrees.

a. 315°

b. 30°

c. 240°

d. 90° ✓

Working through this trigonometry step: $\frac{\pi}{2} \times \frac{180}{\pi} = 90^\circ$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -4$. Find S_{12}, the sum of the first 12 terms.

a. -198

b. -240

c. **-216** ✓

d. -221

$S_n = \frac{n}{2}(2u_1 + (n-1)d) \Rightarrow S_{12} = \frac{12}{2}(2(4) + (12-1)(-4)) = -216.$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 5 blue balls, 5 green balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{12}$

b. $\frac{1}{2}$

c. $\frac{5}{12}$ ✓

d. $\frac{1}{6}$

Working through this statistics step: $P(\text{blue}) = \frac{5}{12} = \frac{5}{12}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Geometry & Trigonometry

Use the change of base formula to evaluate $\log_2(13)$ correct to 2 decimal places.

a. 4.00

b. **3.70** ✓

c. 3.40

d. 3.10

Working through this algebra step: $\log_2(13) = \frac{\ln 13}{\ln 2} \approx 3.70$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Statistics & Probability

A bag contains 6 yellow balls, 3 blue balls, and 5 red balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{7}$

b. $\frac{9}{14}$

c. $\frac{3}{14}$

d. $\frac{5}{14}$ ✓

Working through this statistics step: $P(\text{red}) = \frac{5}{14} = \frac{5}{14}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Calculus

A bag contains 5 blue balls, 4 yellow balls, and 8 red balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{5}{17}$

b. $\frac{9}{17}$

c. $\frac{4}{17}$

d. $\frac{8}{17}$ ✓

Working through this statistics step: $P(\text{red}) = \frac{8}{17} = \frac{8}{17}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number & Algebra

A bag contains 8 blue balls, 6 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{5}{19}$

b. $\frac{6}{19}$ ✓

c. $\frac{8}{19}$

d. $\frac{13}{19}$

Working through this statistics step: $P(\text{red}) = \frac{6}{19} = \frac{6}{19}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Functions

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{4}{9}$

b. $\frac{2}{9}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{3}$

Working through this statistics step: $P(\text{blue}) = \frac{3}{9} = \frac{1}{3}$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Geometry & Trigonometry

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{1}{2}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$ ✓

d. $\frac{3}{4}$

Working through this statistics step: $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Statistics & Probability

A bag contains 3 red balls, 3 green balls, and 2 blue balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{3}{4}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{8}$

Working through this statistics step: $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Calculus

A bag contains 8 blue balls, 4 red balls, and 2 green balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{1}{7}$

b. $\frac{4}{7}$

c. $\frac{2}{7}$ ✓

d. $\frac{5}{7}$

Working through this statistics step: $P(\text{red}) = \frac{4}{14} = \frac{2}{7}$.

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